# 原文佐證：經顱光生物調節（tPBMT）與犬認知功能障礙症候群（CCDS）

建議方案所依據的書中段落，直接以原文截圖呈現，旁邊附中文重點；詳細討論之後補上。

來源

Winkler CJ, Miller LA (eds.). Laser Therapy in Veterinary Medicine: Photobiomodulation. 2nd ed. Wiley; 2025. Ch.16 Neurological Conditions.（書頁 pp.200–202）

使用限制

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# 1　作用原理（病理與 PBMT 機轉）

原文截圖　書 p.200 右欄

dysfunction, inflammation, oxidative brain damage, and deposition of β-amyloid (Aβ) around blood vessels and neurons. Over time, these factors lead to progressive loss of dendrites, synapses, and neurons, followed by cognitive decline (Landsberg *et al.,* 2012; Chapagain *et al.,* 2018; Dewey *et al.,* 2019). As for AD in humans, there is no cure, although a variety of supplements, pharmacological agents, and environmental strategies have been used to treat CCDS, including L-deprenyl, S-Adenosyl methionine, diets fortified with antioxidants, mitochondrial cofactors, and essential fatty acids (Landsberg *et al.,* 2011; Manteca, 2011). While these methods may be helpful, authors and colleagues have observed that the use of transcranial PBMT seems to be very effective in improving the symptoms of CCDS within 4–6 weeks in most cases.

What is interesting from a PBMT standpoint is that compromised blood flow, mitochondrial dysfunction, oxidative stress, and inflammation are at the core of this disease process, and these all happen to be where PBMT shines as a treatment modality, having been extensively discussed in this and other chapters. Vascular compromise and inflammation result in the accumulation of neurotoxic Aβ, generating the production of ROS and proinflammatory cytokines (e.g., IL-1β, IL-6, and TNF-α), stimulating premature neuronal apoptosis via various signalling pathways, and increasing formaldehyde levels (FA) in the brain, all of which can lead to cellular and neuronal damage (De Taboada *et al.,* 2011; Da Luz Eltchechem *et al.,* 2017; Hamblin, 2019). While neuronal mitochondria are the primary source of energy production in the brain, when functioning normally they can also mitigate the toxic effects of Aβ on neuronal cellular function. Compromised cerebral blood flow and the toxic effects of Aβ contribute to mitochondrial dysfunction (Landsberg *et al.,* 2012; Chapagain *et al.,* 2018; Dewey *et al.,* 2019), which subsequently worsens the adverse effects of Aβ on cellular and mitochondrial function in a vicious cycle (Gonzalez-Lima *et al.,* 2014; Da Luz Eltchechem *et al.,* 2017; De La Torre, 2017; Lu *et al.,* 2017).

Studies have demonstrated the ability of PBMT to cause disaggregation of Aβ via suppression of β-secretase activity and stimulation of enzymes responsible for degrading Aβ peptides (De Taboada *et al.,* 2011; Da Luz Eltchechem *et al.,* 2017; Yue *et al.,* 2019; Zhang *et al.,* 2020; Tao *et al.,* 2021). Microglia play a central role in the generation and dissolution of Aβ in rodent AD models. PBMT has been shown to favor a shift between the two main types of activated microglia to the neuroprotective and anti-inflammatory M2 phenotype (Hamblin, 2019; Tao *et al.,* 2021). Rodent studies have demonstrated that tPBMT application decreased brain levels of Aβ, improved cognitive test scores, increased cerebral vascular density, and decreased FA (Huang *et al.,* 2020; Tao *et al.,* 2021).

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## 中文重點

- CCDS 的病理是多因子的：血管受損、粒線體功能障礙、發炎、氧化壓力、Aβ 沉積。
- 作者與同儕觀察到 tPBMT 在多數病例 4–6 週內改善症狀（作者觀察，不是對照試驗）。
- 核心邏輯：血流、粒線體、氧化壓力、發炎，正是 PBMT 擅長處理的地方。
- 血管受損與發炎 → Aβ 堆積 → 活性氧（ROS）與促發炎細胞激素↑、神經元凋亡、甲醛↑，並與粒線體功能障礙互相加重（惡性循環）。
- PBMT 可使 Aβ 解聚、促使微膠細胞偏向 M2 型；囓齒類研究見 Aβ 下降、認知測試改善、腦血管密度增加、甲醛下降。

**討論（待補）**

之後補上。

# 2 證據等級

原文截圖　書 p.201 左欄上段

While there are no veterinary studies published regarding the efficacy of use of tPBMT for the treatment of CCDS, one publication has discussed the pathophysiology and previously mentioned mechanisms by which PBMT may be beneficial (Dewey *et al.,* 2022). The ideal parameters for use of tPBMT for CCDS are not well established. However, data extrapolated from rodent and human clinical trials involving AD (Barrett and Gonzalez-Lima, 2013; Naeser *et al.,* 2014; Saltmarche *et al.,* 2017; Chan *et al.,* 2019; Salehpour *et al.,* 2019; Spera *et al.,* 2021) may be used as a basis of where to start. Though numerous wave-

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## 中文重點

- **沒有已發表的獸醫研究**探討 tPBMT 治療 CCDS 的療效。
- **理想參數尚未確立**；可從嚙齒類與人類阿茲海默症試驗外推。
- 660、810、980、1064 nm 都被研究過，但穿透力不同，犬顱骨上方肌肉比嚙齒類與人類厚。

**討論（待補）**

之後補上。

# 3　劑量、接觸方式、輻照度與總能量

原文截圖　書 p.201 右欄

In a recent literature review examining PBM to prevent and treat AD, authors Enengl, Hamblin, and Dunkl (Enengl *et al.,* 2020) suggested that based on the available body of research on PBM to the brain (which took into consideration both safety and efficacy), application should achieve a dose in the minimum effective range between 5 and 10 J/cm$^2$ at the level of the cortex. Based on the exponential energy decrease during transcranial application, energy densities of 20–60 J/cm$^2$ applied to the human scalp would be suggested and should be somewhat similar in large to very large dogs. The same authors suggest that pulsing with alpha or gamma frequencies might add additional benefit to tPBM against AD, and a 50% duty cycle would necessitate a doubling of the calculated treatment time to achieve the same dose as with CW light.

The authors recommend that the handpiece be held on-contact with the dorsal surface of the skull, treating both the right and left hemispheres (Figures 16.4, 16.5), when treating CCDS. Laser safe eyewear for the patient is recommended and is discussed further in Chapters 4 and 5. When treating with a higher power laser (or at the higher end of the irradiance range listed below), the laser operator is encouraged to keep the treatment head in continuous motion at all times. A handheld class 3b laser may be held in place and used to treat point-to-point as an alternative. An irradiance between 0.250 and 1.5 W/cm$^2$ at the skin surface is recommended. The resultant irradiances at the cortical surface in dogs should replicate dosing that was safe and effective in humans (Ilic *et al.,* 2006) and tested in other species. Therefore, the laser operator must know the spot area of their laser handpiece as well as the power setting and duty cycle (if pulsed) for their laser to adequately calculate dose. In a medium- to large-size dog, treating over the dorsal calvarium, the size area may be between 50 and 70 cm$^2$ which would result in a total dose of 1000–1400 total joules delivered over the entire area if using

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## 中文重點

- **皮質層**最低有效劑量 5–10 J/cm²（Enengl 等人 2020，人類文獻回顧）。
- 經顱能量呈指數衰減，人類頭皮建議 **20–60 J/cm²**，大型到超大型犬可相似。
- alpha / gamma 脈衝可能有額外益處；**50% duty cycle 時間要加倍**。
- 手持件**接觸**顱骨背側，**左右半球**都治療；建議為病患配戴護目鏡。
- 高功率或輻照度在高端時，治療頭要**全程持續移動**；三級 B 雷射可逐點定位。
- 皮膚表面輻照度 **0.250–1.5 W/cm²**；必須知道光斑面積、功率與 duty cycle 才能計算劑量。
- 中大型犬頭頂 50–70 cm²，以 20 J/cm² 計算，總劑量約 **1000–1400 J**（下一段接續）。

**討論（待補）**

之後補上。

# 4 療程頻率與耐受性

原文截圖　書 p.202 左欄

a fluence of 20 J/cm$^2$ (scale down for smaller patients and treatment size areas accordingly). As with most chronic, degenerative conditions, PBMT should be started with 2–3 sessions the first week, if possible, weaning to a transition phase of less-frequent treatments (twice weekly, then once weekly), and continued once every 2–3 weeks as needed, keeping clinical signs at a minimum. Treatments are painless and well tolerated. Some colleagues have noted that the patient may seem a little tired for the first 24 hours, but resolution of clinical signs usually begins within the first one or two treatments and gets progressively better in the first month (Godine, 2017). Bimonthly maintenance treatments offer the optimal chance for continued efficacy Figure 16.6.

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## 中文重點

- 接續上一段：以 20 J/cm$^2$ 計算，**較小的病患依面積縮小**。
- 慢性退化性疾病：**第 1 週 2–3 次**，之後減為每週 2 次、每週 1 次，再**每 2–3 週**維持，以症狀最小化為準。
- 治療**無痛**、**耐受良好**；部分病患前 24 小時可能稍微疲倦。
- 症狀通常在前 1–2 次治療內開始緩解，第一個月逐步進步。
- **每兩個月**的維持治療最有機會持續維持療效。

**討論（待補）**

之後補上。

# 5　不要用的情況

原文截圖　書 p.202 右欄上段（創傷性腦損傷一節）

Animal-model studies using rats have demonstrated that transcranial near-infrared laser application may provide benefit in cases of acute traumatic brain injuries (Xuan *et al.*, 2013). The same parameters as used for treating CCDS can be used to treat traumatic brain injury. The frequency of treatment should be every 12–24 hours as needed for aiding in resolution of clinical signs. PBMT can decrease intracranial inflammation, decrease oxidative stress, and help stabilize damaged membranes. It is not recommended that subacute head trauma cases be treated with PBMT due to the possibility of intracranial hemorrhage; PBMT may cause a transient increase in blood flow. If there is any reason to suspect that intracranial bleeding is ongoing, PBMT should not be used.

書 p.202 右欄上段（創傷性腦損傷一節）

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## 中文重點

- 創傷性腦損傷可使用與 CCDS 相同的參數，頻率每 12–24 小時一次。
- **不建議亞急性頭部外傷使用**：PBMT 可能造成血流短暫增加，有顱內出血的風險。
- **只要有理由懷疑顱內持續出血，就不應使用 PBMT。**

**討論（待補）**

之後補上。

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